

Human Obstacle Detector and Tracker for Acme Robotics

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1. OVERVIEW & OBJECTIVES	2. APPROACH & METHODOLOGY
Project: Human Obstacle Detector and Tracker Duration: 3 weeks Core Objectives: • Detect humans in real-time • Track persistent IDs for each human • Estimate 3D position in robot-centric frame • Deliver a TDD module with 90%+ code coverage	Development Process: • Agile Iterative Process (AIP) • Test-Driven Development (TDD) • Pair Programming with role rotation • CI/CD with GitHub Actions & CodeCov
3. TECHNICAL DETAILS	4. SCHEDULE & DELIVERABLES
Technologies: • C++17, CMake, OpenCV 4.x • Google Test (GTest), cppcheck, Doxygen • ONNX Runtime (for DNN inference) • GitHub Actions (CI), CodeCov (Coverage) Key Algorithms: • DNN-based 2D Detection (YOLOv8) • Intersection-over-Union (IoU) Tracking • Monocular Depth Estimation (MiDaS) • 2D-to-3D Pinhole Camera Projection	Project Schedule: • Wk 1 (Phase 0): Design, UML, Documentation • Wk 2 (Phase 1): Core Implementation (Sprints 1-2) • Wk 3 (Phase 2): Integration & Testing (Sprint 3) Key Deliverables: ✓ UML diagrams & design documentation ✓ CI/CD pipeline with 90%+ test coverage ✓ Production-ready C++ module ✓ Doxygen-generated developer API ✓ Final demonstration video